L1 $$s = (s_1, \dots, s_k) \in \mathbb{F}_p^k, \qquad s_i < 2^{120}, \qquad p = 2^{127} - 1$$

L2 $$\tau = x^{k+2} + \sum_{i=1}^{k} s_i x^i, \qquad a = (s_1, \dots, s_k, x, \tau)$$

L3 $$\Sigma = C\,V, \qquad V = \left[\, j^m \,\right]_{0 \le m < t,\ 1 \le j \le n}$$

L4 $$\sigma = \langle L, R \rangle = \sum_{i=1}^{\ell} L_i R_i, \qquad (L, R) \in \mathbb{F}_p^{\ell} \times \mathbb{F}_p^{\ell}, \quad \ell = 8$$

L5 $$A^{\top} = -A \implies \langle AR, R \rangle = 0, \qquad L \leftarrow L + AR, \quad R \leftarrow R + BL, \quad L \leftarrow L + CR$$

L6 $$\hat{a}_c = \sum_{i=1}^{\ell} \sum_{j \in T} (\lambda_j L_{j,c,i})\, R_{j,c,i}, \qquad \lambda_j = \prod_{m \in T \setminus \{j\}} \frac{m}{m - j}$$

Leakage $$\mathrm{TV}(\Lambda_\sigma, \Lambda_{\sigma'}) \le \left(1 + \tfrac{1}{p}\right) 2^{(\lambda_L + \lambda_R)/2}\, p^{-(\ell-3)/2}$$

Cross-epoch $$\mathrm{TV}\Big((R_{\mathrm{old}}, L_{\mathrm{new}})_\sigma, (R_{\mathrm{old}}, L_{\mathrm{new}})_{\sigma'}\Big) = p^{-(\ell-1)} \approx 2^{-889}$$

AMD bound $$\Pr_x\left[\tau + g = f(x + e, s + d)\right] \le \frac{k+1}{p} \approx 2^{-125}$$